Week 1 Homework

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Exercise 1 - Time Series:

1. Extract the monthly retail trade and food services data –not seasonally adjusted—from 1992 to 2026 from the website below. Save the file in csv form (you need to clean up the dataset in Microsoft Excel) and read it in R studio. Draw time series plot with the data. Attach the plot. Describe types of fluctuations that are observed from the data. Is there any trend observed over time? Any seasonal variations? Irregular fluctuation?

Read and Inspect the Data

```
data_retail<-read.csv("retail.csv")

#head(data_retail)
#tail(data_retail)
#class(data_retail)
```

Observe Value variable is character type

```
class(data_retail$Value)
```

```
## [1] "character"
```

Convert Value to Numeric

```
ourdata<-as.numeric(gsub(",", "", data_retail$Value))
head(ourdata)
```

```
## [1] 142051 142498 154439 158430 164788 163417
```

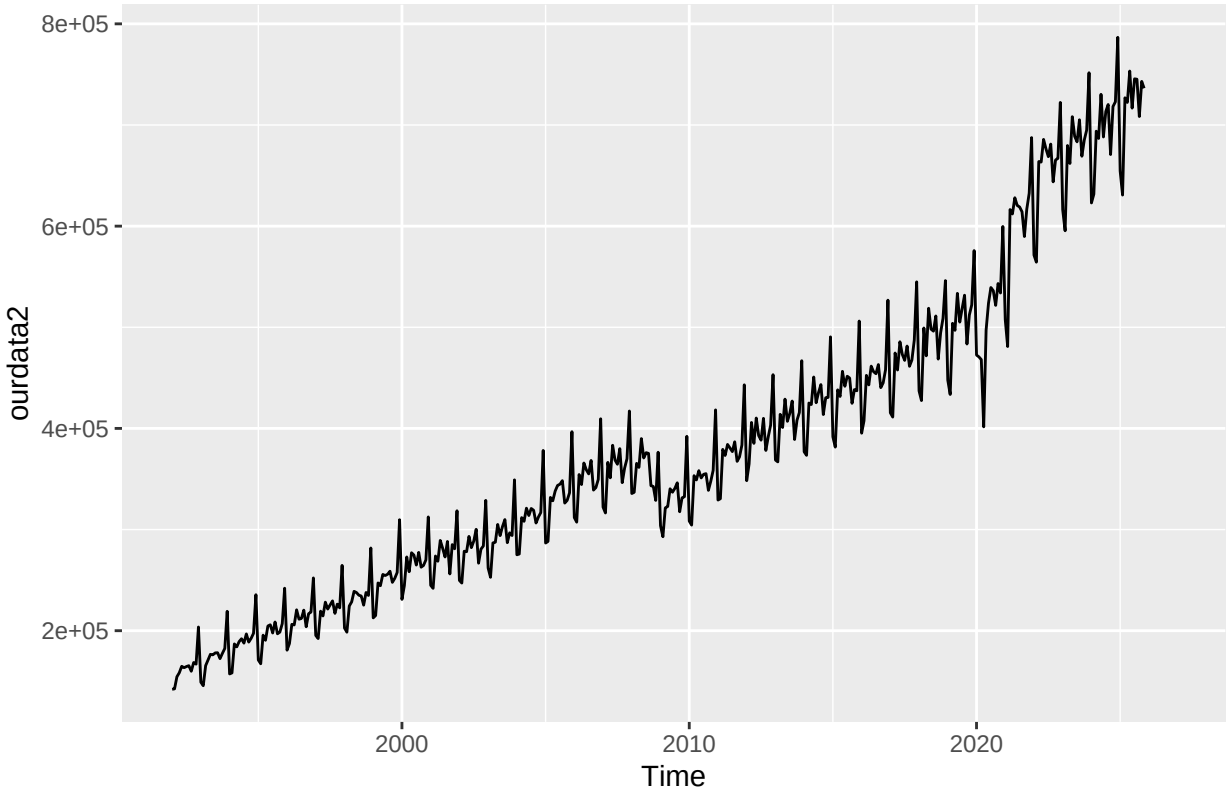
Create Time Series Type

```
ourdata2 <- ts(ourdata, start=c(1992, 1), frequency = 12)
head(ourdata2)
```

```
##         Jan    Feb    Mar    Apr    May    Jun
## 1992 142051 142498 154439 158430 164788 163417
```

Plot

```
autoplot(ourdata2)
```



Exercise 1 Response:

1. There is a general and consisent upward trend from 1992 - 2026

2. There is a slight dip in 2010 and 2020, likely due to the housing marking crash and COVID-19 respectively, but the upward trend resumes after that.

Exercise 2 - Regression:

1. Use the forward selection method with "ames" data, select a model based on BIC, and compare the model fit using BIC, AIC to the one we obtained with the backward method.
2. Use Cp and forward/backward selection methods to choose models with "ames" data and compare the model fit to the ones you obtained with BIC.

Read in the Data

```
ames <- modeldata::ames
```

BIC Backward

```
regfit <- regsubsets(Sale_Price ~ ., data = ames, method = "backward", nvmax = 150)
```

```
regfit.s <- summary(regfit)
dim(regfit.s$outmat)
```

```
## [1] 151 276
```

```
bic_best <- which.min(regfit.s$bic)
bic_best
```

```
## [1] 82
```

```
fullmodel<-model.matrix(Sale_Price ~ ., data = ames)
P<-as.data.frame(fullmodel)
D<-cbind(Sale_Price=ames$Sale_Price,P[,-1])
colnames(D) <- gsub("&","and",colnames(D))
# choose the 82 predictors
fs <- names(which(regfit.s$outmat[bic_best, ] == "*"))
fs <- gsub("&","and",fs)
selected_fs <- paste(fs, collapse = "+")
paste("Sale_Price ~", selected_fs)
```

```
formula_regsubset <- as.formula(paste("Sale_Price ~", selected_fs))
```

```
model_bic <- lm(formula_regsubset, data = D)
#check the results
BIC(model_bic)
```

```
## [1] 68005.86
```

```
AIC(model_bic)
```

```
## [1] 67503.31
```

BIC Forward

```
regfit2 <- regsubsets(Sale_Price ~ ., data = ames, method = "forward", nvmax = 150)
```

```
regfit.s2 <- summary(regfit2)
dim(regfit.s2$outmat)
```

```
## [1] 151 276
```

```r
bic_best2 <- which.min(regfit.s2$bic)
bic_best2
```

```
## [1] 85
```

```r
fullmodel2<-model.matrix(Sale_Price ~ ., data = ames)
P2<-as.data.frame(fullmodel2)
D2<-cbind(Sale_Price=ames$Sale_Price,P2[,-1])
colnames(D2) <- gsub("&","and",colnames(D2))
# choose the 85 predictors
fs2 <- names(which(regfit.s2$outmat[bic_best2, ] == "*"))
fs2 <- gsub("&","and",fs2)
selected_fs2 <- paste(fs2, collapse = "+")
paste("Sale_Price ~", selected_fs2)
```

```r
formula_regsubset2 <- as.formula(paste("Sale_Price ~", selected_fs2))
```

```r
model_bic2 <- lm(formula_regsubset2,data=D2)
#check the results
BIC(model_bic2)
```

```
## [1] 68015.1
```

```r
AIC(model_bic2)
```

```
## [1] 67494.6
```

CMP Backward

```r
regfit3 <- regsubsets(Sale_Price ~ ., data = ames, method = "backward", nvmax = 150)
```

```r
regfit.s3 <- summary(regfit3)
dim(regfit.s3$outmat)
```

```
## [1] 151 276
```

```r
bic_best3 <- which.min(regfit.s3$cp)
bic_best3
```

```
## [1] 135
```

```r
fullmodel3<-model.matrix(Sale_Price ~ ., data = ames)
P3<-as.data.frame(fullmodel3)
D3<-cbind(Sale_Price=ames$Sale_Price,P3[,-1])
colnames(D3) <- gsub("&","and",colnames(D3))
# choose the 135 predictors
fs3 <- names(which(regfit.s3$outmat[bic_best3, ] == "*"))
fs3 <- gsub("&","and",fs3)
selected_fs3 <- paste(fs3, collapse = "+")
paste("Sale_Price ~", selected_fs3)
```

```r
formula_regsubset3 <- as.formula(paste("Sale_Price ~", selected_fs3))
```

```r
model_cp3 <- lm(formula_regsubset3,data=D3)
#check the results
BIC(model_cp3)
```

```
## [1] 68224.03
```

```r
AIC(model_cp3)
```

```
## [1] 67404.39
```

CMP Forward

```r
regfit4 <- regsubsets(Sale_Price ~ ., data = ames, method = "forward", nvmax = 150)
```

```r
regfit.s4 <- summary(regfit4)
dim(regfit.s4$outmat)
```

```
## [1] 151 276
```

```r
bic_best4 <- which.min(regfit.s4$cp)
bic_best4
```

```
## [1] 135
```

```r
fullmodel4 <-model.matrix(Sale_Price ~ ., data = ames)
P4<-as.data.frame(fullmodel4)
D4<-cbind(Sale_Price=ames$Sale_Price,P4[,-1])
colnames(D4) <- gsub("&","and",colnames(D4))
# choose the 135 predictors
fs4 <- names(which(regfit.s4$outmat[bic_best4, ] == "*"))
fs4 <- gsub("&","and",fs4)
selected_fs4 <- paste(fs4, collapse = "+")
paste("Sale_Price ~", selected_fs4)
```

```r
formula_regsubset4 <- as.formula(paste("Sale_Price ~", selected_fs4))
```

```r
model_cp4 <- lm(formula_regsubset4,data=D4)
#check the results
BIC(model_cp4)
```

```
## [1] 68221.74
```

```r
AIC(model_cp4)
```

```
## [1] 67402.1
```

```r
print("BIC Backward")
BIC(model_bic)
AIC(model_bic)

print("BIC Forward")
BIC(model_bic2)
AIC(model_bic2)

print("CMP Backward")
BIC(model_cp3)
AIC(model_cp3)

print("CMP Forward")
BIC(model_cp4)
AIC(model_cp4)
```

Exercise 2 Response:

1. For the BIC Backwards model (model select 82 variables), the BIC was 68005.86 and AIC was 67503.31 whereas for the BIC Forward model (model selected 85 variables), the BIC was 68015.1 and the AIC was 67494.6. BIC selected the Backwards model and AIC selected the forward model. I would go with the Backwards model here since BIC penalized more strictly for the inclusion of more variables to prevent over fitting.

2. For the CMP Backward model (model selected 135 variables), the BIC was 68224.03 and the AIC was 67404.39 whereas the CMP Forward model (model selected 135 variables), the BIC was 68221.74 and the AIC was 67402.1. Each model included 135 variables, and both BIC and AIC agree that the forward model was the stronger model.